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Guidelines for Structural Health Monitoring

Suresh Bhalla¹, Naveet Kaur^{2*} and Rohit Gogna³

¹Professor, Department of Civil Engineering, Indian Institute of Technology (IIT), President of ISHMS, Delhi, India

²Vice President ISHMS, New Delhi, India

³Founder and Chairman, Sanrachna Prahri Private Limited, General Secretary of ISHMS, India

***Corresponding author:** Naveet Kaur, Vice President ISHMS, New Delhi, India

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Abstract

The Indian Structural Health Monitoring Society (ISHMS) has introduced the Guidelines for Structural Health Monitoring (SHM) to address the urgent need for systematic monitoring and maintenance of India's rapidly growing infrastructure. Approved in the 4th General Body Meeting on 10 June 2024, this document provides a foundational framework for implementing SHM across critical structures, ensuring informed decision-making through data-driven approaches. The guidelines recommend SHM implementation from the construction phase or at the earliest for existing structures, targeting high-rise buildings, specific types of bridges, nuclear power plants, vibration-sensitive structures, and large-span domes or spaces. Key parameters to be monitored include static and dynamic metrics such as deflection, strain, temperature, and natural frequencies, with a focus on deploying advanced sensor technologies like vibrating wire strain gauges, fiber-optic sensors, and piezoelectric sensors. Instrumentation modes range from manual to IoT-enabled systems, with a preference for automated, real-time data acquisition and analysis. The guidelines emphasize regular reporting, threshold-based alert generation, and feature extraction to assess structural conditions effectively. Specific structural components such as foundations, bridge decks, and building joints are prioritized for instrumentation to ensure compliance with national codes like IRC 112 and IS 1893. This document serves as a comprehensive guide for stakeholders to adopt modern SHM practices, reducing downtime, optimizing maintenance, and enhancing the longevity and safety of infrastructure.

Keywords: SHM; Guidelines; bridges; Concrete structures

Abbreviation: SHM: Structural Health Monitoring; RTF: Run to Failure; GC: Governing Council; GCM: Governing Council Meeting; ISHMS: Indian Structural Health Monitoring Society; PZT: Lead zirconate titanate; CVS: Concrete vibration sensors; MFC: Macro-Fiber composite; IoT: internet of things

Introduction

India is witnessing a high surge in construction activities as part of the infrastructure boost accompanying its strong economic growth. Numerous bridges, tall buildings, airports, railway stations and similar structures are currently under various stages of

construction all over the country. Unfortunately, not enough seems to be done with regard to the monitoring and maintenance of the ever-growing infrastructure repository. There is no code of practice at the present related to structural health monitoring (SHM) in India. The Indian infrastructure industry typically follows the "run



to failure (RTF)" model as far maintenance is concerned. This means ignoring any monitoring or maintenance needs till some problem creeps in. No sensor-based data driven scientific approach is often followed. SHM is defined as a continuous sensor-based data driven approach aimed to facilitate informed decision about current health condition of the structure. This is the right time Indian infrastructure industry adopts modern SHM sensors and related technologies so that decisions backed by scientific data are arrived at any future instant of time rather than relying on the RTF approach leading to greater downtime as well as higher recovery cost for infrastructure. This document is prepared by the Governing Council (GC) of the Indian Structural Health Monitoring Society (ISHMS) based on its body of expert members and also after receiving feedback from industry experts as well as relevant government agencies. The second General Body Meeting (GBM) held on 11 Dec

2023 decided to prepare a document serving as basic guideline to various stakeholders. The Governing Council Meeting (GCM) held on 12 January 2024 decided to receive suggestions from industry also. Accordingly, online surveys were conducted from 06 March 2024 to receive inputs from industry and other stakeholders. The document was approved in the Fourth GBM held on 10 June 2024. The document attempts to provide basic guidelines to stakeholders to implement SHM in their structures.

Structures on which should SHM be implemented?

It is recommended that SHM be implemented on category of structures mentioned in Table 1 right from the construction stage. In case of existing structures, it is recommended to implement SHM at the earliest possible:

Table 1: Structures where it is recommended to implement SHM.

Type of Structure	Recommendation
Buildings	> 15 storeys or height 50 m and above: up to zone IV (as per IS1893) > 7 storeys or height 20 m and above: zone V (as per IS1893)
Bridges	(i) RCC bridges with span equal to or exceeding 25 m
	(ii) Prestressed girder bridges with span equal to or exceeding 30 m for I-type 40 m for Box type
	(iii) Steel truss bridges with span equal to or exceeding 75 m
	(iv) All cable stayed, suspension and extradosed bridges irrespective of span
	(v) Skew bridges with span equal to or exceeding 20 m: RCC I-shaped girder, 25 m: Pre-stressed I-shaped girders 25 m: RCC Box-type girder, 30 m: Prestress Box-type girder
Nuclear power plants	
Structures under continuous vibrations, such as critical machine foundations, such as those for turbo generators	
Structures with importance factor of more than one as per IS 1893 (I)	
Spaces/ domes exceeding 30 m span	

Parameters To Be Monitored as Part of SHM

Generally, SHM involves monitoring static and dynamic parameters. Concerned agencies may decide to instrument sensors for measuring following parameters:

a. **Static:** Deflection, strain, temperature

b. **Dynamic:** Deflection, strain, at least first natural frequency and mode shape

Sensors

Sensors play crucial role in the process of SHM. Table 2 provides the list of commercially available sensors, which the concerned agencies might consider while instrumenting a structure

Table 2: List of commercially available sensors for SHM.

1.	Vibrating wire strain gauges
2.	Linear Variable Displacement Sensor (LVDT)
3.	Laser based sensor/ techniques
4.	Tilt/ inclinometers (for deflections)
5.	Piezo sensors (for global vibration technique or electro-mechanical impedance) - Lead zirconate titanate (PZT) for surface bonding - Concrete vibration sensors (CVS) for embedding inside RC structures - Macro-fiber composite (MFC) for surface bonding

6.	Accelerometers
7.	Vision based techniques/ sensors
8.	Fiber-optic sensors
9.	Any other scientifically established sensors deemed relevant to the structure or the component

Instrumentation Cum Measurement Mode

Depending upon the expertise and budget available, concerned agencies may decide to adopt either manual or internet of things (IoT) based automatic mode (preferred) for data acquisition from the instrumented sensors. If IoT based instrumentation is adopted, the data should preferably be available to users in online real-time graphical user interface. Suitable sampling interval be adopted for static and dynamic measurement such that it is able to enable appropriate feature extraction, whether static strain/ deflection or dynamic parameters such as natural frequencies, mode shapes or any structural signature representative of the condition of the

structure. Sampling should be accompanied by a report generation, which should be at least monthly for IoT based monitoring and at least six monthly for manual SHM. The measured data should be analysed to enable suitable feature extraction based on the particular measurement and utilizing the currently available SHM literature. In case of IoT mode, the system should be capable of generating alert/ warning in case of threshold exceedance.

Structural Components to be Instrumented

It is recommended to instrument the components of structures mentioned in Table 3 [1-3].

Table 3: List of commercially available sensors for SHM.

S.No.	Structural Component	Parameters to be measured
1.	Foundations/ piles	Settlement, inclination, tilt etc.
2.	Bridge-deck	Strain (static/ dynamic), deflection and acceleration. Static strains are relatively easy to measure as compared to deflection and can be used for compliance to deflection limits imposed by bridge codes such as IRC 112 (2020). Following theoretical expressions (based on Euler-Bernoulli beam theory) can be used to convert measured mid-span strain into mid-span deflection for simply supported bridges: - Under uniformly distributed load: $\delta = \frac{5L^2}{24D} \epsilon$ - Under point load: $\delta = \frac{L^2}{3D} \epsilon$
3.	Bridge Piers and abutments	
4.	Beams, columns and joints of framed structures	
5.	Building floors for acceleration response measurement under wind/ earthquake [refer IS 16700 (2017)]	

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References

1. IRC 112 (2020) Code of Practice for Concrete Road Bridges.
2. IS 16700 (2017) Criteria for Structural Safety of Tall Concrete Buildings.
3. IS 1893 (I) Criteria for Earthquake Resistant Design of Structures: Part 1 General Provisions and Buildings.